**Full Stack Assignment – 20**

**Assignment ID:** FSD-20

**Title:** Feedback Collection Management System

**Duration:** 3 Hours

## 1️⃣ Objective

Design and develop a **Feedback Collection Management System** that allows users to submit feedback and administrators to review it.

The system must enforce a time-based edit restriction and ensure **data persistence using a database**.

## 2️⃣ User Roles

**USER**

- Register and log in to the system
- Submit feedback
- View submitted feedback
- Edit feedback within allowed time window

**ADMIN**

- Log in to the system
- View all feedback submissions

## 3️⃣ Functional Requirements

**Authentication & Authorization**

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

**Feedback Management**

- USER can submit feedback with required details
- Feedback must be stored in the database
- ADMIN can view all feedback entries
- Feedback data must be retrieved dynamically from the database

## 4️⃣ Business Rule (Critical)

### ⚠️ Feedback can be edited only within 15 minutes of submission.

Any edit attempt after the time limit must be rejected at the backend.

## 5️⃣ Technical Constraints (Mandatory)

- All feedback data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

## 6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Feedback submission form
- Feedback listing with edit option (time-restricted)
- Read-only feedback view for ADMIN

UI design is not evaluated; time-based logic and correctness matter.

## 7️⃣ Submission Requirements (Strict)

**GitHub Repository**

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

**Live Deployment**

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

**README.md (Mandatory)**

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

### 8️⃣ Evaluation Note

Evaluation focuses on **time-based rule enforcement, role control, and end-to-end execution**.